1064 Center Dr
Gainesville, FL 32611
March 13, 2026

Dr. Kurt Schulze
University of Florida
1064 Center Dr
Gainesville, FL 32611

Dear Dr. Schulze:

The objective of this experiment is to characterize the performance of the departmental double pipe heat exchanger for potential research applications. Specifically, the Log Mean Temperature Difference (LMTD) will be evaluated for both parallel and counter-flow configurations, and the overall heat transfer coefficient area product (UA) is going to be determined experimentally. These values will be compared to theoretical predictions obtained using empirical heat transfer correlations. Additionally, the fouling factor will be estimated, axial temperature distributions will be analyzed, and the uncertainty in UA for each flow configuration is going to be quantified to establish the thermal capability and reliability of the apparatus for future research use.

The analysis was based on conservation of energy applied to a steady-flow control volume encompassing the heat exchanger. First, the Logarithmic Mean Temperature Difference (LMTD) method was used to model heat transfer by calculating the effective average temperature difference between the hot and cold-water flows. Heat transfer rate was then found by multiplying volumetric flow rate by the density and isobaric specific heat, both of which varied with flow temperature, and by the absolute value of the outlet temperature minus the inlet temperature. Through these experimental results, the overall heat transfer coefficient area product (UA) was determined by dividing the average heat transfer rate between the hot and cold stream by the LMTD temperature for both parallel and counter flow configurations at equal and unequal flow rates. These values were compared to theoretical predictions for UA obtained using empirical heat transfer correlations involving Reynolds number, Prandtl number, and Nusselt number and a thermal resistance analysis from the hot fluid in the inner pipe to the cold fluid in the outer pipe. Additionally, the fouling factor, R_f'', was estimated from the predicted and theoretical UA values. The ε-NTU method was employed to determine theoretical temperature distributions along the axial direction of the exchanger through calculations involving the efficiency of the system, flow parameters and exchanger geometry, and the uncertainty in UA was quantified.

The experimental apparatus consisted of a 3.05-meter-long concentric tube heat exchanger with a copper inner pipe and brass outer pipe. Hot water heated with an electric resistance heater flowed through the inner pipe while cold water flowed through the annulus, or the volume between the inner and outer pipe. The hot-water flow was always in the same direction, but a valve arrangement allowed cold water to flow in in either parallel or counterflow configurations. Temperatures at the inlet, midpoint, and exit of both streams were measured with calibrated Omega Model ON-403-PP tubular probe thermistors, and volumetric flow rates were measured using two calibrated rotary flow sensors consisting of a block enclosure and flexible rotor. Tests were conducted for both exchanger configurations at equal flow rates of approximately 4 LPM and at unequal flow rates of 4 LPM and 7 LPM for hot and cold flow, respectively. Steady-state conditions signaled by a green light on the VI interface were established prior to recording all data to ensure thermal equilibrium and measurement consistency.

The experimental results showed similar thermal behavior for both parallel and counterflow configurations. For the counterflow exchanger at equal flow rates, the LMTD was 12.96 °C and produced an experimental UA value of 121.3 W/K compared to a theoretical value of 230.1 W/K. When the cold-water stream flow rate was increased, the LMTD slightly decreased to 12.77 °C while the experimental UA increased to 167.3 W/K, indicating enhanced convective heat transfer due to the higher flow rate. For the parallel flow configuration, the LMTD was 12.49 °C at equal flow rates with an experimental UA of 137.69 W/K compared to a theoretical value of 234.0 W/K. Increasing the cold flow rate resulted in a similar LMTD of 12.25 °C but increased the experimental UA to 182.75 W/K with a corresponding theoretical value of 275.6 W/K, exhibiting similar behavior to what was observed for the counterflow configuration. The fouling factor values for both configurations remained low indicating minimal thermal resistance, and they decreased when flow rate increased because lower fluid velocities produced weaker turbulence near the pipe walls. Across all cases, the predicted temperature distributions obtained from the ε-NTU method closely matched the experimentally measured temperatures, indicating that the analytical model accurately captured the thermal behavior of the heat exchanger. Overall, the consistent heat transfer trends across both flow configurations indicates that the double pipe heat exchanger provides reliable and predictable thermal performance and is therefore suitable for potential research applications.

Sincerely,

Bryce Addeo | Sean Muller | Matthew Strogach

"On my honor, I have neither given nor received unauthorized aid in doing this assignment."

Table 1: Before performing the heat exchanger experiment, the flowmeter had to be calibrated to convert a voltage differential to a flow rate. To do this, hot and cold-water tubes from the heat exchanger were used in conjunction with a five-gallon bucket, graduated cylinder and calibration VI. First, the end of the tube connected to the cold-water line was set in a five-gallon bucket and the hot and cold-water valves were closed to ensure all water flowed through the calibration hose. The calibration VI was turned on to observe the voltage generated by increasing flow rate and the valve was opened one revolution so that water began to flow into the bucket. Once the flow reached steady state, determined by the stability of the voltage reading on the VI, the hose was moved to the graduated cylinder, and a timer was simultaneously started. As the cylinder filled with water, the average voltage was observed on the VI for the given flow rate. Once 1 L of water was filled in the cylinder, the timer was immediately stopped and noted down along with the average voltage value throughout the run. The same process was repeated five more times, with an additional revolution being applied to the valve dial each time so that by the last trial it was opened six revolutions. The cold-water valve was then completely closed and the voltage reading for zero flow rate was noted. This same process was then repeated for the hot water flow resulting in another six data points of voltage, time and volume. To complete the calibration process, the data had to be plotted on Excel to determine the polynomial coefficients shown in the table below. First, the zero voltage values were added to all the voltage data points to account for the offset when there was no flow rate. Next, flow rate was solved by dividing the measured volume, in this case always equal to 1 L, by the time in minutes (converted from the initial measurements in seconds) it took to fill that volume. Finally, flow rate was plotted on the y-axis of a scatter plot against voltage on the x-axis, for both cold and hot flows, and a polynomial trendline was plotted with the equations displayed. The coefficient in front of the x^2 term was A, the coefficient in front of the x^1 term was B and the coefficient in front of the x^0 term was C. The A coefficient controls the curvature of the calibration curve, representing the flowmeter sensitivity at higher voltages. A positive value means the flow rate increases slightly faster as voltage increases, and the value is greater for the hot-water flow meaning the relationship between voltage and flow becomes more nonlinear at higher flow rates. The B coefficient represents the direct proportional relationship between voltage and flow rate, and the higher value for cold flow means that the cold flowmeter had a much stronger linear response between voltage and flow rate. Finally, the C coefficient represents the interception of the calibration curve when voltage is zero and accounts for sensor offset, small measurement bias and noise in the electronics. The larger C value for the hot-water flow suggests a larger offset in the hot flowmeter signal, meaning a non-zero flow rate would be predicted even at very low voltages.

Experimental Unit Location			
Flowmeter Calibration Constants	**A** $(\frac{LPM}{V^2})$	**B** $(\frac{LPM}{V})$	**C** (LPM)
Cold Flow	0.0603	1.231	1.2429
Hot Flow	0.1772	0.3772	3.023

Table 2: This table showcases the hot and cold, inlet and outlet water temperatures measured during the experiment for both parallel and counter flow configurations at equal and unequal flow rates. For the counter flow configuration, both hot and cold-water flow temperatures decreased along the axial direction. The cold water decreased from around 28 °C to about 23 °C for both flow rates, and the hot water decreased from approximately 43 °C to 35-36 °C for both flow rates. Next, for the parallel flow configuration, the hot water flow decreased from roughly the same inlet temperature to similar outlet temperatures, but the cold-water flow now increased in temperature from inlet to outlet. The inlet and outlet cold-water temperatures observed for the counter flow configuration essentially flipped from the parallel flow configuration, which makes logical sense since the direction of cold-water flow is flipped when changing between exchanger configurations and thus the axial temperature distribution follows suit. However, the hot and cold-water temperatures in general remain similar across all cases because the inlet temperatures, heat exchanger geometry, and overall heat transfer conditions remained nearly constant throughout the experiment. Additionally, the hot and cold flow rates were comparable, giving similar heat capacity rates and resulting in roughly the same amount of heat transfer regardless of whether the exchanger operated in parallel or counter flow.

	Cold	**Hot**	**Cold**	**Cold**	**Hot**	**Hot**
	Flow Rate (LPM)	**Flow Rate (LPM)**	$\mathbf{T_{in}}$ $(°C)$	$\mathbf{T_{out}}$ $(°C)$	$\mathbf{T_{in}}$ $(°C)$	$\mathbf{T_{out}}$ $(°C)$
Counter Flow	3.95 ± 0.12	4.03 ± 0.12	28.45 ± 0.05	22.92 ± 0.02	42.39 ± 0.04	36.55 ± 0.03
Counter Flow	6.89 ± 0.21	3.97 ± 0.12	27.50 ± 0.02	23.05 ± 0.01	42.91 ± 0.02	35.11 ± 0.02
Parallel Flow	4.05 ± 0.12	4.04 ± 0.12	23.37 ± 0.04	29.30 ± 0.01	42.98 ± 0.01	36.65 ± 0.02
Parallel Flow	6.92 ± 0.21	3.95 ± 0.12	23.36 ± 0.01	27.96 ± 0.01	43.17 ± 0.01	34.87 ± 0.01

Table 3: This table summarizes all of the flow rates, log mean temperature differences, *LMTD*, experimental heat transfer coefficients, $UA_{experimental}$, theoretical heat transfer coefficients, $UA_{theoretical}$, and fouling factor, R_f'' for the counter-flow and parallel-flow configurations in the heat exchanger. The cold and hot flow rates were both set to 4 LPM for equal flow rates and to 4 LPM and 7 LPM for unequal flow rates. The calculated LMTD values, representing the average temperature difference between the hot and cold water along the entire length of the heat exchanger, did not deviate much. The fact that all the LMTD values did not change much and have small uncertainties verified that the driving force temperature for heat transfer remained relatively consistent. Using those temperature differences, the heat transfer coefficients were able to be calculated and indicate that the experimentally measured heat transfer performance was lower than the ideal prediction. Essentially, this proves that the heat exchanger did not transfer as much heat in the real-life experiment as theory expected. This is because theory does not account for imperfections where experimental models have them. The fouling factors were very small, indicating there was little thermal resistance in the system due to a very small amount of material buildup and turbulence at the walls of the pipes.

	Cold Flow Rate (LPM)	**Hot Flow Rate (LPM)**	**LMTD ($°C$)**	**UA Experimental (W/K)**	**UA Theoretical (W/K)**	**R_f'' ($m^2 * K/W$)**
Counter Flow	3.95 ± 0.12	4.03 ± 0.12	12.96 ± 0.153	121.35 ± 3.94	230.14 ± 1.98	0.00022 ± 1.54 x 10^{-5}
Counter Flow	6.89 ± 0.21	3.97 ± 0.12	12.77 ± 0.138	167.29 ± 5.23	275.17 ± 1.45	0.000132 ± 1.07 x 10^{-5}
Parallel Flow	4.05 ± 0.12	4.04 ± 0.12	12.49 ± 0.117	137.69 ± 4.06	234.01 ± 1.58	0.000169 ± 1.23 x 10^{-5}
Parallel Flow	6.92 ± 0.21	3.95 ± 0.12	12.25 ± 0.108	182.75 ± 5.29	275.55 ± 1.44	0.000104 ± 9.06 x 10^{-6}

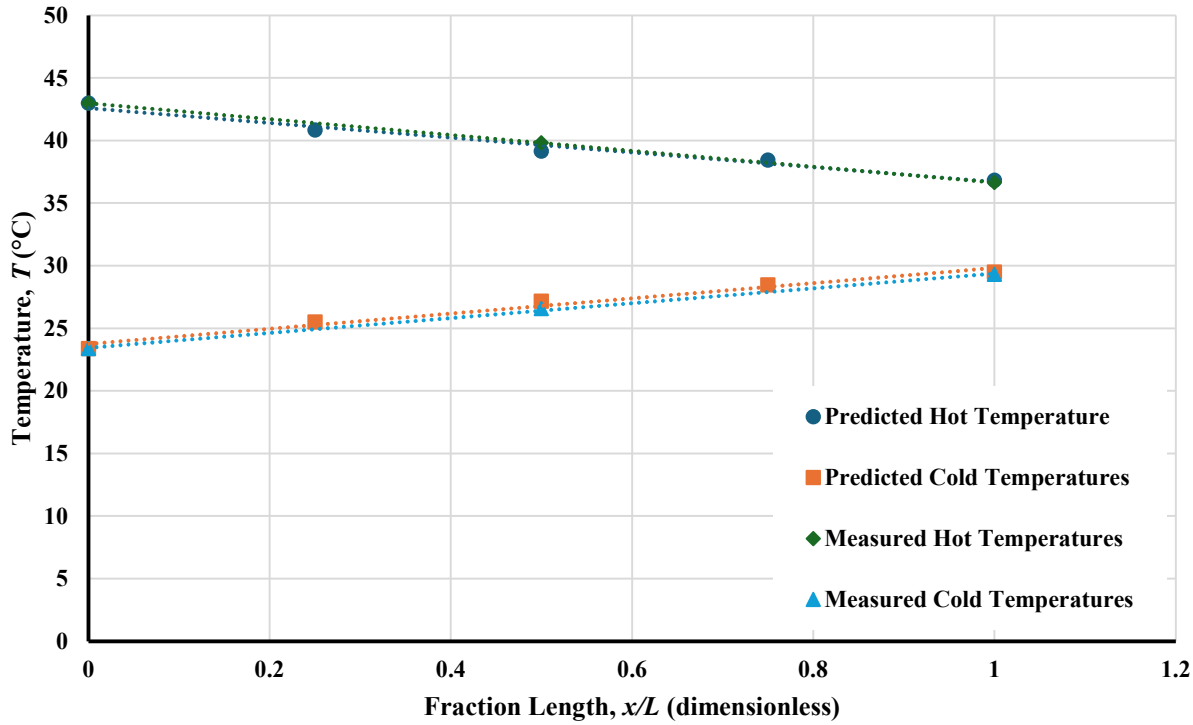

Figure 1: This graph showcases the experimentally measured inlet, midpoint and outlet temperatures and the predicted axial temperature distribution found using the ε-NTU method for both hot and cold flows for the parallel exchanger configuration with equal flow rates. The x-axis of the graph features fraction length which denotes the location of the temperature along the axial length of the pipes, with 0 corresponding to a length of 0 meters and 1 corresponding to the full length of 3.05 meters. From inlet to outlet, temperature linearly decreases for the hot flow and linearly increases for the cold flow. The magnitude of both the decrease in hot temperature and the increase in cold flow temperature is roughly 6.5 °C over the entire pipe length, meaning the rate of change, or slope of the line, is the same for both temperature distributions. Another similarity between both hot and cold flows, is that the predicted temperatures solved from the ε-NTU method are so close in value to the measured values from the experiment that they essentially form equal trend lines. Looking through a critical lens, it can be observed that the predicted temperatures slightly underestimate the hot temperatures and slightly overestimate the cold temperatures. However, these differences are by such a small margin that no useful conclusion can be made outside of the fact that the ε-NTU method produces high accuracy temperature distribution predictions for the experimental heat exchanger. In the parallel heat exchanger, the hot and cold stream temperatures approach each other as shown above but do not fully converge because the temperature difference between the two fluids decreases along the length of the exchanger, which reduces the driving force for heat transfer, and thus the streams asymptotically approach each other but cannot reach the same temperature within a finite exchanger length.

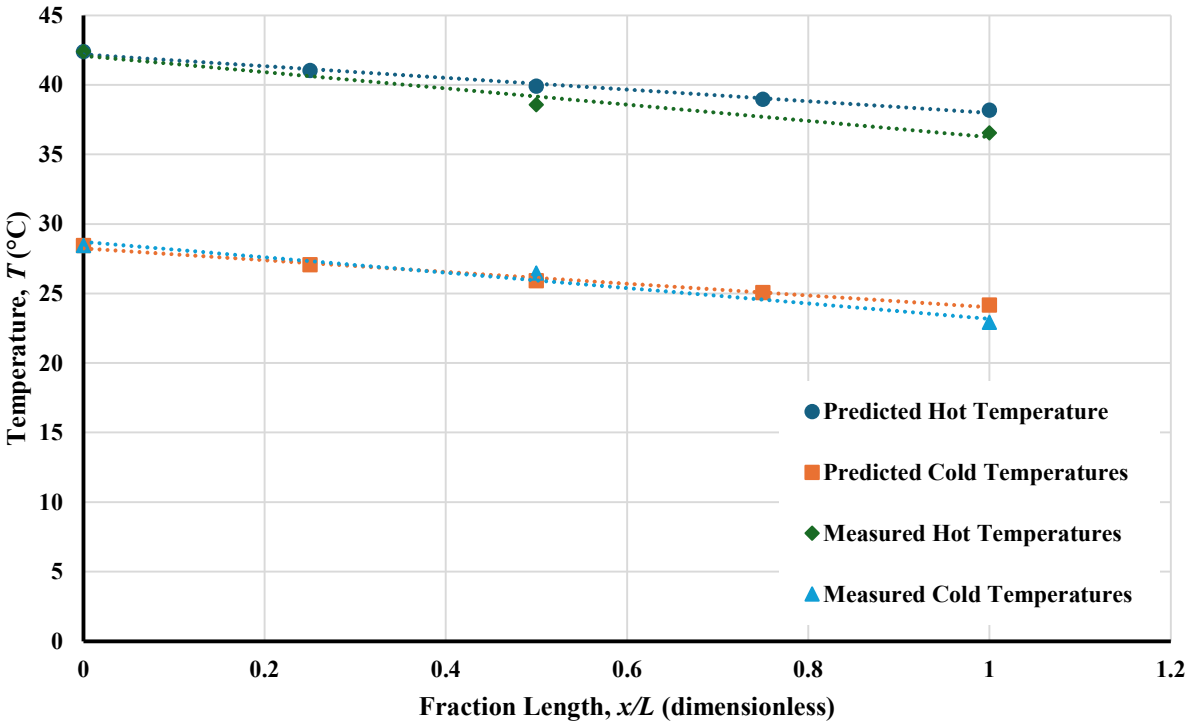

Figure 2: This graph exhibits the experimentally measured inlet, midpoint and outlet temperatures and the predicted axial temperature distribution found using the ε-NTU method for both hot and cold-water streams for the counterflow exchanger configuration with equal flow rates. Similar to the previous graph, the predicted temperature distributions and actual measured temperatures are approximately equal for both hot and cold flows. However, a major difference observed for the counter flow configuration is that both the hot and cold flows decreased in temperature from the inlet to outlet whereas before the cold temperatures increased along the exchanger length. This is because in counterflow heat exchangers the cold fluid flows opposite to the hot fluid. Since the fraction length axis is referenced in the direction of the hot flow, the cold-water temperatures appear to decrease along the pipe even though the cold fluid is still gaining heat as it travels through the exchanger in the opposite direction. Excluding the fact that the inlet and outlet temperatures are flipped for the cold flow, the overall maximum and minimum temperature values are essentially the same between both parallel and counter flow configurations at equal flow rates. While the flow configuration changes how the temperatures vary along the exchanger length, it does not significantly change the total energy exchanged, so the inlet and outlet temperature magnitudes remain nearly the same.

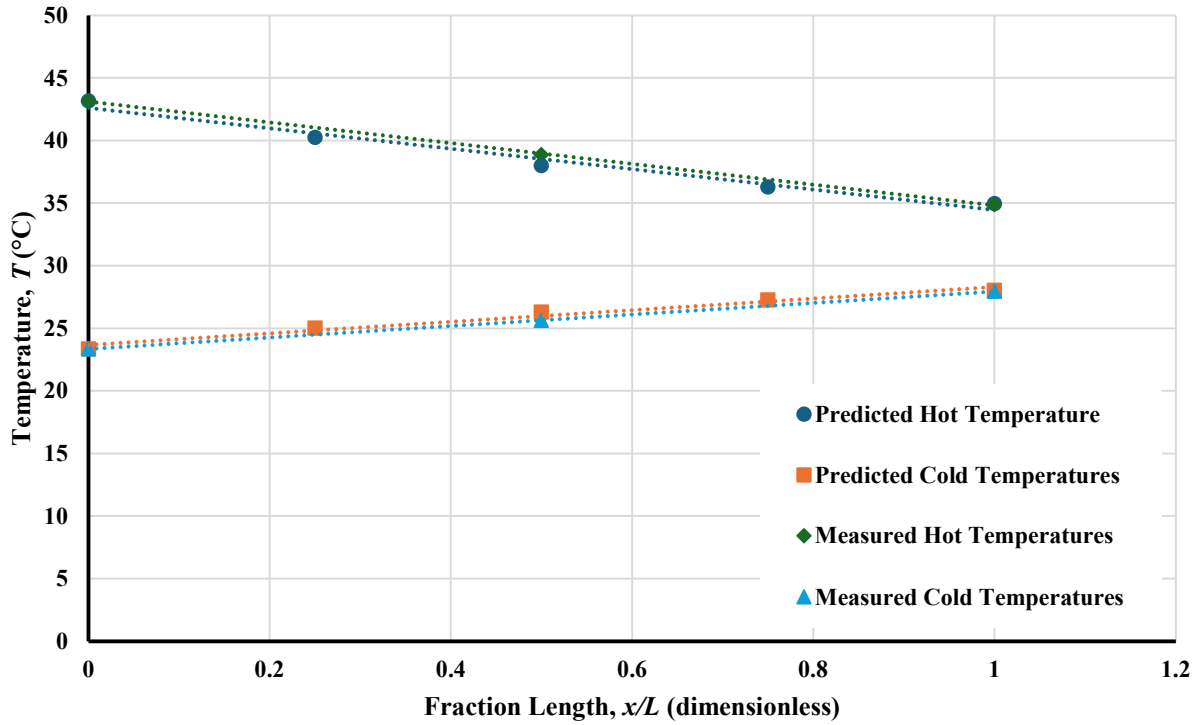

Figure 3: This graph showcases the experimentally measured inlet, midpoint and outlet temperatures and the predicted axial temperature distribution found using the ε-NTU method for both hot and cold flows for the parallel exchanger configuration with unequal flow rates. Whereas before both the hot and cold flows had a volumetric rate of approximately 4 LPM, the cold-water flow rate was now increased to roughly 7 LPM. Comparing these results to those expressed in the graph from Figure 1 for the parallel configuration with equal flow rates reveals no substantial differences in trends and the predicted temperatures still align with measured values. However, upon closer inspection, it can be observed that the outlet temperatures for both the hot and cold flows are about 1.5 °C lower for the unequal flow rates despite the inlet temperatures being about the same as before. This is because increasing the flow rate increases the heat capacity rate of the stream, meaning more fluid mass is carrying the heat. Since the same amount of heat transfer is distributed over a larger mass flow, the temperature change of the fluid becomes smaller, resulting in outlet temperatures that are closer to the inlet values compared to the equal flow rate case.

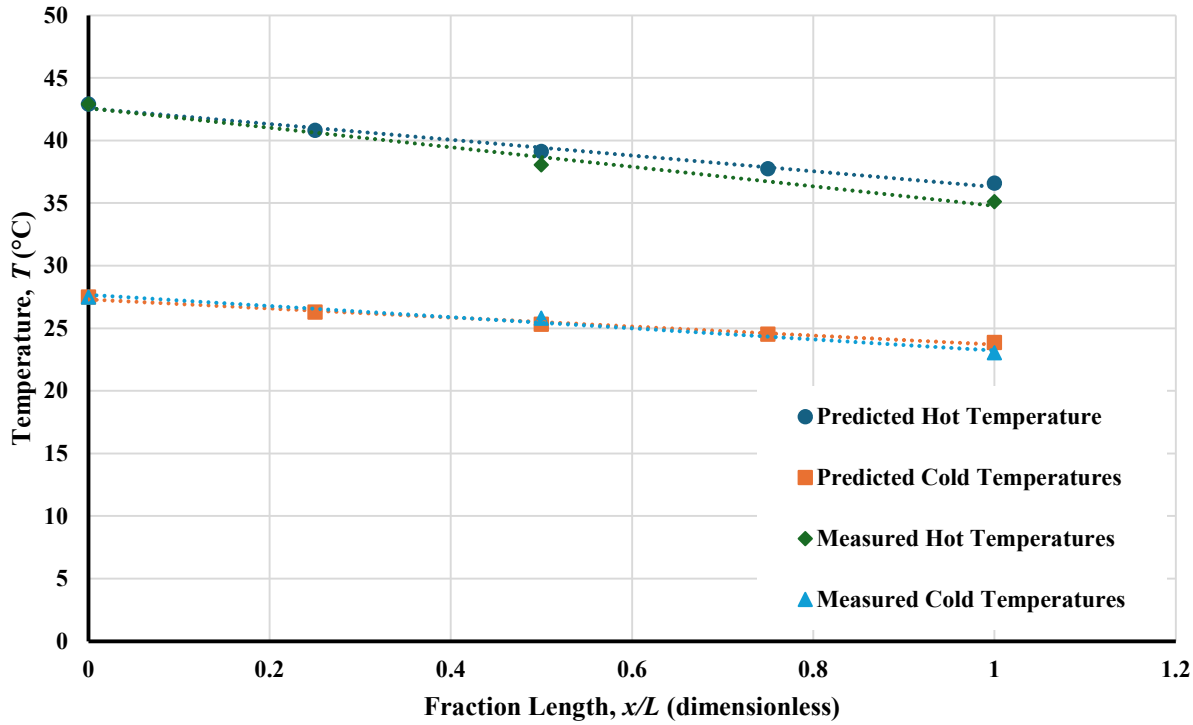

Figure 4: This graph exhibits the experimentally measured inlet, midpoint and outlet temperatures and the predicted axial temperature distribution found using the ε-NTU method for both hot and cold-water streams for the counterflow exchanger configuration with unequal flow rates. The flow rates for the hot and cold streams expressed in the graph above were the same as those detailed in Figure 3. Once again, no differences in trends can be observed in the graph above compared to the previous counterflow configuration test case with equal flow rates shown in Figure 2. There is even less variance between the outlet temperatures than what was observed for the parallel heat exchanger between equal and unequal flow rates. In general, the magnitudes of the temperatures remained similar for all test cases regardless of flow rate or exchanger configuration because the inlet temperatures, heat exchanger geometry and overall heat transfer conditions remained nearly constant throughout the experiment as previously stated when observing the values in Table 2. However, seeing the data plotted graphically further proved this point, especially for the hot water temperature distribution which remained approximately the same for all four test cases.

Table 4: This table shows the uncertainty in $UA_{experimental}$ and $UA_{theoretical}$ for each of the configurations. Additionally, the contribution of each parameter is shown. For experimental uncertainty, it was identified that the heat transfer rate, q_{avg}, and the log mean temperature difference, $LMTD$, had significant impact on the total uncertainty. When determining the influence of each measured factor on the theoretical uncertainty, it was concluded that the length, L, diameters, D_i and D_o, and pipe conductivity, k_p, had negligible influence when compared to the convection coefficients, h_i and h_o, which were influenced by Reynolds number and Prandtl number. As such, they were not included in this table. For both the counter flow and parallel flow configurations, two flow conditions were used. Equal flow rates, which used a flow rate of 4 LPM for both the hot and cold flows, and unequal flow rates, which used a flow rate of 4 LPM for the hot flow and 7 LPM for the cold flow.

Configuration	$\mu_{UA_{ex}}$	$\frac{\partial UA_{ex}}{\partial q_{avg}}\mu_{q_{avg}}$	$\frac{\partial UA_{ex}}{\partial LMTD}\mu_{LMTD}$	$\mu_{UA_{th}}$	$\frac{\partial UA_{th}}{\partial h_i}\mu_{h_i}$	$\frac{\partial UA_{th}}{\partial h_o}\mu_{h_o}$
Counter Flow, Equal Flow Rates	3.964	3.942	0.421	1.978	1.572	1.120
Counter Flow, Unequal Flow Rates	5.238	5.227	0.331	2.268	1.449	1.744
Parallel Flow, Equal Flow Rates	4.056	4.049	0.225	2.006	1.584	1.230
Parallel Flow, Unequal Flow Rates	5.290	5.289	0.114	2.270	1.444	1.751
Units	$\left[\pm\frac{W}{°C}\right]$	$\left[\frac{W}{°C}\right]$	$\left[\frac{W}{°C}\right]$	$\left[\pm\frac{W}{°C}\right]$	$\left[\frac{W}{°C}\right]$	$\left[\frac{W}{°C}\right]$

The following sample calculation solves for the Log Mean Temperature Difference, *LMTD*, which gives a temperature difference to represent the average driving force for heat transfer across the heat exchanger. The following calculation is for a counterflow example with uneven flowrates, where the cold flowrate is 7 LPM and the hot flowrate is 4 LPM. The log mean temperature difference is based on the temperature difference at one end of the heat exchanger, ΔT_1, and the temperature difference at the other end of the heat exchanger, ΔT_2.

$$LMTD = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \; p.11 \; Schulze \; [1] \tag{1}$$

The temperature differences are derived from subtracting the hot and cold fluid temperatures that are entering and leaving the heat exchanger for both configurations. Different variables are subtracted depending on if the heat exchanger configuration is counterflow or parallel. The four temperature difference options come from the two temperature differences at each end of the heat exchanger for counter flow, $\Delta T_{1,counter}$ and $\Delta T_{2,counter}$ and another two for parallel flow, $\Delta T_{1,parallel}$ and $\Delta T_{2,parallel}$.

$$\Delta T_{1,counter} = T_{hot,inlet} - T_{cold,outlet} \; p.15 \; Schulze \; [2] \tag{2}$$

$$\Delta T_{2,counter} = T_{hot,outlet} - T_{cold,inlet} \; p.15 \; Schulze \; [2] \tag{3}$$

$$\Delta T_{1,parallel} = T_{hot,inlet} - T_{cold,inlet} \; p.15 \; Schulze \; [2] \tag{4}$$

$$\Delta T_{1,parallel} = T_{hot,outlet} - T_{cold,outlet} \; p.15 \; Schulze \; [2] \tag{5}$$

Since an uneven counterflow example was selected for this calculation, the temperature differences were derived from (2) and (3), and then the LMTD was solved.

$$\Delta T_{1,counter} = 42.9°C - 23.1°C = 19.8°C$$

$$\Delta T_{2,counter} = 35.1°C - 27.5°C = 7.60°C$$

$$LMTD = \frac{\Delta T_{1,counter} - \Delta T_{2,counter}}{\ln\left(\frac{\Delta T_{1,counter}}{\Delta T_{2,counter}}\right)}$$

$$LMTD = \frac{19.8°C - 7.60°C}{\ln\left(\frac{19.8°\cancel{C}}{7.60°\cancel{C}}\right)}$$

$$\therefore LMTD = 12.7°C$$

The uncertainty in the log mean temperature difference was found using RSS propagation where the partial derivative of the LMTD equation was taken with respect to each of its variables. Each partial derivative was multiplied by the uncertainty of the variable that each derivative was taken with respect to. The uncertainties of the temperature difference at one end of the heat exchanger,

$\Delta T_{1,counter}$, and at the other end of the heat exchanger, $\Delta T_{2,counter}$ were also found through RSS propagation.

$$\mu_{\Delta T_{1,counter}} = \sqrt{(\mu_{T_{hot,inlet}} \frac{\delta \Delta T_{1,counter}}{\delta T_{hot,inlet}})^2 + (\mu_{T_{cold,outlet}} \frac{\delta \Delta T_{1,counter}}{\delta T_{cold,outlet}})^2}$$

$$\frac{\delta \Delta T_{1,counter}}{\delta T_{hot,inlet}} = \frac{\delta T_{ho\ ,inlet} - T_{cold,outlet}}{\delta T_{hot,inlet}} = 1$$

$$\frac{\delta \Delta T_{1,counter}}{\delta T_{cold,outlet}} = \frac{\delta T_{hot,inlet} - T_{cold,outlet}}{\delta T_{cold,outlet}} = -1$$

$$\mu_{T_{hot,inlet}} = 1.96 * std\ deviation + 0.1 = 1.96 * 0.00993 + 0.1 = 0.119°C$$

$$\mu_{T_{cold,outlet}} = 1.96 * std\ deviation + 0.1 = 1.96 * 0.00546 + 0.1 = 0.111°C$$

$$\mu_{\Delta T_{1,counter}} = \sqrt{(0.119°C * 1)^2 + (0.111°C * -1)^2} = 0.163°C$$

$$\mu_{\Delta T_{2,counter}} = \sqrt{(\mu_{T_{ho\ ,outlet}} \frac{\delta \Delta T_{2,counter}}{\delta T_{hot,outlet}})^2 + (\mu_{T_{cold,inlet}} \frac{\delta \Delta T_{2,counter}}{\delta T_{cold,inlet}})^2}$$

$$\frac{\delta \Delta T_{2,counter}}{\delta T_{ho\ ,outlet}} = \frac{\delta T_{hot,outlet} - T_{cold,inlet}}{\delta T_{ho\ ,outlet}} = 1$$

$$\frac{\delta \Delta T_{2,counter}}{\delta T_{cold,inlet}} = \frac{\delta T_{hot,outlet} - T_{cold,inlet}}{\delta T_{cold,inlet}} = -1$$

$$\mu_{T_{ho\ ,outlet}} = 1.96 * std\ deviation + 0.1 = 1.96 * 0.00986\ °C + 0.1\ °C = 0.119°C$$

$$\mu_{T_{cold,inlet}} = 1.96 * std\ deviation + 0.1 = 1.96 * 0.0142\ °C + 0.1\ °C = 0.128°C$$

$$\mu_{\Delta T_{2,counter}} = \sqrt{(0.119°C * 1)^2 + (0.128°C * -1)^2} = 0.175°C$$

$$\mu_{LMTD} = \sqrt{(\mu_{\Delta T_{1,counter}} \frac{\delta LMTD}{\delta \Delta T_{1,counter}})^2 + (\mu_{\Delta T_{2,counter}} \frac{\delta LMTD}{\delta \Delta T_{2,counter}})^2}$$

$$\frac{\delta LMTD}{\delta \Delta T_{1,counter}} = \frac{\ln\left(\frac{\Delta T_{1,counter}}{\Delta T_{2,counter}}\right) - \frac{\Delta T_{1,counter} - \Delta T_{2,counter}}{\Delta T_{1,counter}}}{\ln\left(\frac{\Delta T_{1,counter}}{\Delta T_{2,counter}}\right)^2} = \frac{\ln\left(\frac{19.8°\cancel{C}}{7.6°\cancel{C}}\right) - \frac{19.8°C - 7.6°C}{19.8°\cancel{C}}}{\ln\left(\frac{19.8°\cancel{C}}{7.6°\cancel{C}}\right)^2} = 0.372°C$$

$$\frac{\delta LMTD}{\delta \Delta T_{2,counter}} = -\frac{\ln\left(\frac{\Delta T_{1,counter}}{\Delta T_{2,counter}}\right) + \frac{\Delta T_{1,counter} - \Delta T_{2,counter}}{\Delta T_{2,counter}}}{\ln\left(\frac{\Delta T_{1,counter}}{\Delta T_{2,counter}}\right)^2} = -\frac{\ln\left(\frac{19.8°\cancel{C}}{7.6°\cancel{C}}\right) + \frac{19.8°C - 7.6°C}{7.6°\cancel{C}}}{\ln\left(\frac{19.8°\cancel{C}}{7.6°\cancel{C}}\right)^2} = 0.707°C$$

$$\mu_{LMTD} = \sqrt{(0.372°C * 0.163°C)^2 + (0.707°C * 0.175°C)^2}$$

$$\therefore\ \mu_{LMTD} = \pm\ 0.138\ °C$$

The following sample calculation solves for the total heat transfer rate, q, which represents the amount of thermal energy transferred from the hot water to the cold water per unit time in the heat exchanger. The total heat transfer rate is dependent on the fluid volumetric flow rate, Q, the fluid density, ρ, the fluid specific heat capacity, C_p, the fluid temperature at the outlet of the heat exchanger, T_o, and the fluid temperature at the inlet of the heat exchanger, T_i.

$$q = Q * \rho * C_p * |T_o - T_i| \; p.14 \; Schulze \; [2] \tag{6}$$

The performed sample calculation for the total heat transfer rate was for the counterflow configuration where the cold flow rate is 7 LPM and the hot flow rate is 4 LPM. To account for both the hot and cold flow heat transfer rates, the average total heat transfer rate, q_{avg}, was determined. In an ideal world the hot heat transfer rate, q_{hot}, and the cold heat transfer rate, q_{cold}, should be the same, but this is not always the case due to small uncertainties and heat loss to surroundings. Therefore, the average is solved to gauge a more accurate estimate.

$$Q_{ho} \;=\; 3.97 \; \cancel{LPM} * \frac{10^{-3} m^3}{\cancel{L}} * \frac{1 \; \cancel{min}}{60 \; s} = 0.0000662 \; \frac{m^3}{s}$$

$$\rho = 992.7 \; \frac{kg}{m^3}$$

$$C_p = 4179.7 \; \frac{J}{Kg * °C}$$

$$T_o = 35.1 \; °C$$

$$T_i = 42.9 \; °C$$

$$q_{ho} \;= 0.0000662 \; \frac{\cancel{m^3}}{s} * 992.7 \; \frac{\cancel{kg}}{\cancel{m^3}} * 4179.7 \; \frac{J}{\cancel{Kg * °C}} * |35.1 \; \cancel{°C} - 42.9 \; \cancel{°C}| = 2142.4 \; W$$

$$Q_{cold} \;=\; 6.89 \; \cancel{LPM} * \frac{10^{-3} m^3}{\cancel{L}} * \frac{1 \; \cancel{min}}{60 \; s} = 0.000115 \; \frac{m^3}{s}$$

$$\rho = 996.9 \; \frac{kg}{m^3}$$

$$C_p = 4181.5 \; \frac{J}{Kg * °C}$$

$$T_o = 23.1 \; °C$$

$$T_i = 27.5 \; °C$$

$$q_{cold} = 0.000115 \; \frac{\cancel{m^3}}{s} * 996.9 \; \frac{\cancel{kg}}{\cancel{m^3}} * 4181.5 \; \frac{J}{\cancel{Kg * °C}} * |23.1 \; \cancel{°C} - 27.5 \; \cancel{°C}| = 2130.6 \; W$$

$$q_{avg} = \frac{q_{hot} + q_{cold}}{2} \tag{7}$$

$$q_{avg} = \frac{2142.4 \; W + 2130.6 \; W}{2}$$

$$\therefore\ q_{avg} = 2136.5\ W$$

The uncertainty in the total heat transfer rate was found using RSS propagation. In order to do this calculation, the partial derivative of the total heat transfer rate equation was taken with respect to each of its variables and was multiplied by the variable it was taken with respect to. Then, the square root of the sum of those squares was taken. To find the uncertainty in the average total heat transfer rate, q_{avg}, the uncertainty in the cold and hot total heat transfer rates had to be taken first. This was also done through RSS propagation. The uncertainty in density, ρ, was found to be 0.05% of the actual measured value [3]. The uncertainty in the specific heat capacity, C_p, was found to be 1% of the actual measured value. The uncertainty in the volumetric flow rate, Q, was found to be 3% of the actual measured value [4]. The temperature uncertainties were found using by assuming an infinite data set and multiplying the standard deviation of each temperature by 1.96.

$$\mu_{q_{hot}} = \sqrt{(\mu_Q \frac{\delta q_{hot}}{\delta Q})^2 + (\mu_\rho \frac{\delta q_{hot}}{\delta \rho})^2 + (\mu_{C_p} \frac{\delta q_{hot}}{\delta C_p})^2 + (\mu_{T_0} \frac{\delta q_{hot}}{\delta T_0})^2 + (\mu_{T_i} \frac{\delta q_{hot}}{\delta T_i})^2} \tag{8}$$

$$\mu_{q_{hot}} = \sqrt{\begin{gathered}\left(\mu_Q * \rho C_p * |T_0 - T_i|\right)^2 + \left(\mu_\rho * Q C_p * |T_0 - T_i|\right)^2 + \left(\mu_{C_p} * Q\rho * |T_0 - T_i|\right)^2 \\ + (\mu_{T_0} * Q\rho C_p)^2 + (\mu_{T_i} * -Q\rho C_p)^2\end{gathered}}$$

$$\mu_{q_{hot}} = \sqrt{\begin{gathered}\left(0.00000199\ \frac{\cancel{m^3}}{s} * 992.7\ \frac{\cancel{kg}}{\cancel{m^3}} * 4179.7\ \frac{J}{\cancel{Kg * °C}} * |35.1\cancel{°C} - 42.9\cancel{°C}|\right)^2 \\ + \left(0.496\ \frac{\cancel{m^3}}{s} * 0.0000662\ \frac{\cancel{m^3}}{s} * 4179.7\ \frac{J}{Kg * °C} * |35.1\cancel{°C} - 42.9\cancel{°C}|\right)^2 \\ + \left(41.8\ \frac{J}{\cancel{Kg * °C}} * 0.0000662\ \frac{\cancel{m^3}}{s} * 992.7\ \frac{\cancel{kg}}{\cancel{m^3}} * |35.1\cancel{°C} - 42.9\cancel{°C}|\right)^2 \\ + \left(0.199\cancel{°C} * 0.0000662\ \frac{\cancel{m^3}}{s} * 992.7\ \frac{\cancel{kg}}{\cancel{m^3}} * 4179.7\ \frac{J}{\cancel{Kg * °C}}\right)^2 \\ + (0.199\cancel{°C} * -0.0000662\ \frac{\cancel{m^3}}{s} * 992.7\ \frac{\cancel{kg}}{\cancel{m^3}} * 4179.7\ \frac{J}{\cancel{Kg * °C}})^2\end{gathered}}$$

$$\mu_{q_{hot}} = 82.095\ W$$

$$\mu_{q_{cold}} = \sqrt{(\mu_Q \frac{\delta q_{cold}}{\delta Q})^2 + (\mu_\rho \frac{\delta q_{cold}}{\delta \rho})^2 + (\mu_{C_p} \frac{\delta q_{cold}}{\delta C_p})^2 + (\mu_{T_0} \frac{\delta q_{cold}}{\delta T_0})^2 + (\mu_{T_i} \frac{\delta q_{cold}}{\delta T_i})^2} \tag{9}$$

$$\mu_{q_{cold}} = \sqrt{\begin{gathered}\left(\mu_Q * \rho C_p * |T_0 - T_i|\right)^2 + \left(\mu_\rho * Q C_p * |T_0 - T_i|\right)^2 + \left(\mu_{C_p} * Q\rho * |T_0 - T_i|\right)^2 \\ + (\mu_{T_0} * Q\rho C_p)^2 + (\mu_{T_i} * -Q\rho C_p)^2\end{gathered}}$$

$$\mu_{q_{cold}} = \sqrt{\begin{gathered}\left(0.00000345\ \frac{\cancel{m^3}}{s} * 996.9\ \frac{\cancel{kg}}{\cancel{m^3}} * 4181.5\ \frac{J}{\cancel{Kg * °C}} * |23.1\cancel{°C} - 27.5\cancel{°C}|\right)^2 \\ + \left(0.498\ \frac{\cancel{m^3}}{s} * 0.000115\ \frac{\cancel{m^3}}{s} * 4181.5\ \frac{J}{Kg * °C} * |23.1\cancel{°C} - 27.5\cancel{°C}|\right)^2 \\ + \left(41.8\ \frac{J}{\cancel{Kg * °C}} * 0.000115\ \frac{\cancel{m^3}}{s} * 996.9\ \frac{\cancel{kg}}{\cancel{m^3}} * |23.1\cancel{°C} - 27.5\cancel{°C}|\right)^2 \\ + \left(0.111\cancel{°C} * 0.000115\ \frac{\cancel{m^3}}{s} * 996.9\ \frac{\cancel{kg}}{\cancel{m^3}} * 4181.5\ \frac{J}{Kg * °C}\right)^2 \\ + (0.128\cancel{°C} * -0.000115\ \frac{\cancel{m^3}}{s} * 996.9\ \frac{\cancel{kg}}{\cancel{m^3}} * 4181.5\ \frac{J}{Kg * °C})^2\end{gathered}}$$

$$\mu_{q_{cold}} = 105.29\ W$$

$$\mu_{q_{avg}} = \sqrt{(\mu_{q_{hot}} \frac{\delta q_{avg}}{\delta q_{ho}})^2 + (\mu_{q_{cold}} \frac{\delta q_{avg}}{\delta q_{cold}})^2} \tag{10}$$

$$\mu_{q_{avg}} = \sqrt{(\mu_{q_{hot}} * \frac{1}{2})^2 + (\mu_{q_{cold}} * \frac{1}{2})^2}$$

$$\mu_{q_{avg}} = \sqrt{(82.095\ W * \frac{1}{2})^2 + (105.29\ W * \frac{1}{2})^2}$$

$$\therefore\ \mu_{q_{avg}} = 66.759\ W$$

The experimental overall heat transfer coefficient area product, $UA_{experimental}$, represents the performance of the heat exchanger as measured through the internal area of the pipes, where the total rate of heat transfer, q_{avg}, was divided by the log mean temperature difference, $LMTD$.

$$UA_{experimental} = \frac{q_{avg}}{LMTD} \; p.3 \; Schulze \; [1] \tag{11}$$

The sample calculation performed for the experimental overall heat transfer coefficient area product was for a counterflow configuration where the cold flow rate is 7 LPM and the hot flow rate is 4 LPM.

$$UA_{experimental} = \frac{2136.5 \; W}{12.7 \; ℃}$$

$$\therefore UA_{experimental} = 167.3 \; \frac{W}{℃}$$

The uncertainty in the experimental overall heat transfer coefficient area product, $\mu_{UA_{experimental}}$, was found using RSS propagation where the partial derivative of the experimental overall heat transfer coefficient area product equation was taken with respect to each of its variables. Each partial derivative was multiplied by the uncertainty of the variable that each derivative was taken with respect to. The uncertainties of the total rate of heat transfer, $\mu_{q_{avg}}$, and the log mean temperature difference, μ_{LMTD}, were both found with RSS propagation.

$$\mu_{UA_{experimental}} = \sqrt{\left(\mu_{q_{avg}} \frac{\partial UA_{experimental}}{\partial q_{avg}}\right)^2 + \left(\mu_{LMTD} \frac{\partial UA_{experimental}}{\partial LMTD}\right)^2} \tag{12}$$

$$\mu_{UA_{experimental}} = \sqrt{\left(\mu_{q_{avg}} \frac{\partial UA_{experimental}}{\partial q_{avg}}\right)^2 + \left(\mu_{LMTD} \frac{\partial UA_{experimental}}{\partial LMTD}\right)^2}$$

$$\mu_{UA_{experimental}} = \sqrt{\left(66.7 \; \cancel{W} \frac{167.3 \frac{W}{℃}}{2136.5 \; \cancel{W}}\right)^2 + \left(0.0253 \; \cancel{℃} \frac{167.3 \frac{W}{℃}}{12.7 \; \cancel{℃}}\right)^2}$$

$$\therefore \mu_{UA_{experimental}} = \pm \, 5.2 \; \frac{W}{℃}$$

The theoretical overall heat transfer coefficient area product, $UA_{theoretical}$, was calculated through the Reynolds number, Re, Prandtl number, Pr, Nusselt number, Nu, and coefficient of convection, h. Additionally, the pipe area, A, pipe diameter, D, pipe length, L, and thermal conductivity of the pipe, k_p, account for the geometry and material properties of the pipe.

$$UA_{theoretical} = \left(\frac{1}{h_i * A_i} + \frac{\ln\left(\frac{D_o}{D_i}\right)}{2 * \pi * k_p * L} + \frac{1}{h_o * A_o}\right)^{-1} \; p.9 \; Schulze \; [2] \tag{13}$$

The Reynolds number was calculated for the inner and outer pipe from the dynamic viscosity, μ, hydraulic diameter, D_h, density, ρ, and the volumetric flow rate, Q.

$$Re = \frac{4 * Q * \rho}{\mu * D_h * \pi} \; Eqn. 7.11, p. 212 \; Fox \; and \; McDonald \; [5] \tag{14}$$

The Prandtl number was calculated from the specific heat capacity, C_p, the dynamic viscosity, and thermal conductivity of the fluid, k_f.

$$Pr = \frac{C_p * \mu}{k_f} \; Eqn. 6.44, p. 299 \; Lienhard \; [6] \tag{15}$$

The Nusselt number was calculated from the Reynolds number, Prandtl number, and the Dittus-Boelter correlation, n. The value of the Dittus-Boelter correlation was 0.4 for heating and 0.3 for cooling, accounting for changes in the viscosity of the water.

$$Nu = 0.023 * Re^{\frac{4}{5}} * Pr^n \; Eqn. 7.39b, p. 368 \; Lienhard \; [6] \tag{16}$$

The heat transfer coefficient, h, was calculated from the Nusselt number, thermal conductivity of the fluid, and hydraulic diameter.

$$h = \frac{Nu * k_f}{D_h} \; Eqn. 6.44, p. 298 \; Lienhard \; [6] \tag{17}$$

The sample calculation performed for the theoretical overall heat transfer coefficient area product was for a counterflow configuration where the cold flow rate is 7 LPM and the hot flow rate is 4 LPM.

$$Re_o = \frac{4 * \frac{3.97}{60000} \frac{\cancel{L}}{\cancel{min}} * \frac{\cancel{m^3}}{\cancel{L}} * \frac{\cancel{min}}{\cancel{s}} * 992.7 \frac{\cancel{kg}}{\cancel{m^3}}}{0.000672 \; \cancel{Pa} * \cancel{s} * 0.0109 \; \cancel{m} * \pi} = 11423.5$$

$$Re_i = \frac{4 * \frac{6.89}{60000} \frac{\cancel{L}}{\cancel{min}} * \frac{\cancel{m^3}}{\cancel{L}} * \frac{\cancel{min}}{\cancel{s}} * 996.9 \frac{\cancel{kg}}{\cancel{m^3}}}{0.000882 \; \cancel{Pa} * \cancel{s} * 0.0097 \; \cancel{m} * \pi} = 17045.0$$

$$Pr_o = \frac{4.17 \frac{\cancel{kJ}}{\cancel{kg} * \cancel{K}} * 1000 \frac{\cancel{J}}{\cancel{kJ}} * 0.000672 \; \cancel{Pa} * \cancel{s}}{0.627 \; \frac{\cancel{W}}{\cancel{m} * \cancel{K}}} = 4.48$$

$$Pr_o = \frac{4.18 \frac{\cancel{kJ}}{\cancel{kg} * \cancel{K}} * 1000 \frac{\cancel{J}}{\cancel{kJ}} * 0.000882 \; \cancel{Pa} * \cancel{s}}{0.607 \; \frac{\cancel{W}}{\cancel{m} * \cancel{K}}} = 6.07$$

$$Nu_o = 0.023 * 11423.5^{\frac{4}{5}} * 4.48^{0.4} = 73.9$$

$$Nu_i = 0.023 * 17045.0^{\frac{4}{5}} * 6.07^{0.3} = 95.9$$

$$h_o = \frac{73.9 * 0.627 \frac{W}{\cancel{m} * K}}{0.0109\ \cancel{m}} = 4247.1 \frac{W}{m^2 * K}$$

$$h_i = \frac{95.9 * 0.607 \frac{W}{\cancel{m} * K}}{0.0097\ \cancel{m}} = 6004.1 \frac{W}{m^2 * K}$$

$$UA_{theoretical} = \left(\frac{1}{6004.1 \frac{W}{\cancel{m^2} * K} * \pi * 3.05\ \cancel{m} * 0.0109\ \cancel{m}} + \frac{\ln\left(\frac{0.0128\ \cancel{m}}{0.0109\ \cancel{m}}\right)}{2 * \pi * 385 \frac{W}{\cancel{m} * K} * 3.05\ \cancel{m}} + \frac{1}{4247.1 \frac{W}{\cancel{m^2} * K} * \pi * 3.05\ \cancel{m} * 0.0128\ \cancel{m}} \right)^{-1}$$

$$\therefore UA_{theoretical} = 275.2 \frac{W}{K} = 275.2 \frac{W}{°C}$$

The following sample calculation solves for the Fouling Factor, R_f'', which represents the additional thermal resistance caused by material buildup on the inside of the heat exchanger pipe that builds up over time. Such build up can be due to corrosion or dirt. The fouling factor is dependent on the experimental heat transfer coefficient, $UA_{experimental}$, the theoretical heat transfer coefficient, $UA_{theoretical}$, the inner heat transfer area, A_i, and the outer heat transfer area, A_o.

$$R_f'' = \frac{\frac{1}{UA_{experimental}} - \frac{1}{UA_{theoretical}}}{\frac{1}{A_i} + \frac{1}{A_o}} \quad Eqn.\ 11.5, p.\ 650\ Bergman\ et\ al.\ [7] \tag{18}$$

This sample calculation was completed for the counterflow configuration with uneven flowrates case, where the cold and hot flow rates were 7 LPM and 4 LPM, respectively.

$$UA_{experimental} = 167.3 \frac{W}{K}$$

$$UA_{theoretical} = 275.2 \frac{W}{K}$$

$$A_i = 275.2\ m^2$$

$$A_o = 0.123\ m^2$$

$$R_f'' = \frac{\frac{1}{167.3 \frac{W}{K}} - \frac{1}{275.2 \frac{W}{K}}}{\frac{1}{0.104\ m^2} + \frac{1}{0.123\ m^2}}$$

$$\therefore R_f'' = 0.000132 \frac{m^2 * K}{W}$$

The following calculation shows the uncertainty of the fouling factor, R_f''. The uncertainty was calculated through RSS propagation where the partial derivative of the fouling factor equation was taken with respect to each variable in the equation. The uncertainty of the experimental heat transfer coefficient, $UA_{experimental}$, the theoretical heat transfer coefficient, $UA_{theoretical}$, the inner heat transfer area, A_i, and the outer heat transfer area, A_o were all found through RSS propagation as well. This specific example uncertainty calculation is for the counterflow configuration with uneven flow rates.

$$\mu_{A_i} = \sqrt{(\mu_{D_{i,i}} \frac{\delta A_i}{\delta D_{i,i}})^2 + (\mu_L \frac{\delta A_i}{\delta L})^2} \tag{19}$$

$$\mu_{A_i} = \sqrt{(\mu_{D_{i,i}} * \pi * L)^2 + (\mu_L * \pi * D_{i,i})^2}$$

$$\mu_{A_i} = 0.0015\, m^2$$

$$\mu_{A_o} = \sqrt{(\mu_{D_{i,o}} \frac{\delta A_o}{\delta Q})^2 + (\mu_L \frac{\delta A_o}{\delta L})^2} \tag{20}$$

$$\mu_{A_o} = \sqrt{(\mu_{D_{i,o}} * \pi * L)^2 + (\mu_L * \pi * D_{i,o})^2}$$

$$\mu_{A_o} = 0.0015\, m^2$$

$$\mu_{UA_{experimental}} = 5.24\ \frac{W}{°C}$$

$$\mu_{UA_{theoretical}} = 2.27\ \frac{W}{°C}$$

$$\mu_{R_f''} = \sqrt{\left(\mu_{UA_{experimental}} \frac{\delta R_f''}{\delta UA_{experimental}}\right)^2 + \left(\mu_{UA_{theoretical}} \frac{\delta R_f''}{\delta UA_{theoretical}}\right)^2 + (\mu_{A_i} \frac{\delta R_f''}{\delta A_i})^2 + (\mu_{A_o} \frac{\delta R_f''}{\delta A_o})^2} \tag{21}$$

$$\mu_{R_f''} = \sqrt{\begin{array}{l} \left(\mu_{UA_{experimental}} * -\dfrac{1}{{UA_{experimental}}^2 * \left(\dfrac{1}{A_i} + \dfrac{1}{A_o}\right)}\right)^2 \\ + \left(\mu_{UA_{theoretica}} * \dfrac{1}{{UA_{theoretica}}^2 * (\dfrac{1}{A_i} + \dfrac{1}{A_o})}\right)^2 \\ + \left(\mu_{A_i} * \dfrac{\left(\dfrac{1}{UA_{experimental}} - \dfrac{1}{UA_{theoretica}}\right)}{{A_i}^2 * \left(\dfrac{1}{A_i} + \dfrac{1}{A_o}\right)}\right)^2 \\ + (\mu_{A_o} * \dfrac{(\dfrac{1}{UA_{experimental}} - \dfrac{1}{UA_{theoretical}})}{{A_o}^2 * (\dfrac{1}{A_i} + \dfrac{1}{A_o})})^2 \end{array}}$$

$$\mu_{R_f''} = \sqrt{\begin{array}{l} \left(5.24\ \dfrac{W}{°C} * -\dfrac{1}{(167.29\ \dfrac{W}{°C})^2 * \left(\dfrac{1}{0.104\ m^2} + \dfrac{1}{0.123\ m^2}\right)}\right)^2 \\ + \left(2.27\ \dfrac{W}{°C} * \dfrac{1}{(275.17\ \dfrac{W}{°C})^2 * (\dfrac{1}{0.104\ m^2} + \dfrac{1}{0.123\ m^2})}\right)^2 \\ + \left(0.0015\ m^2 * \dfrac{\left(\dfrac{1}{167.29\ \dfrac{W}{°C}} - \dfrac{1}{275.17\ \dfrac{W}{°C}}\right)}{(0.104\ \cancel{m^2})^{\cancel{2}} * \left(\dfrac{1}{0.104\ \cancel{m^2}} + \dfrac{1}{0.123\ \cancel{m^2}}\right)}\right)^2 \\ + (0.0015\ m^2 * \dfrac{(\dfrac{1}{167.29\ \dfrac{W}{°C}} - \dfrac{1}{275.17\ \dfrac{W}{°C}})}{(0.123\ \cancel{m^2})^{\cancel{2}} * (\dfrac{1}{0.104\ \cancel{m^2}} + \dfrac{1}{0.123\ \cancel{m^2}})})^2 \end{array}}$$

$$\therefore\ \mu_{R_f''} = 1.07\ \times 10^{-5}\ \frac{m^2 K}{W}$$

This sample calculation solves for the predicted temperature of the hot fluid at the midpoint location of the pipe, $T_{h,0.5}$, and cold fluid at the midpoint location of the pipe, $T_{c,0.5}$, in the heat exchanger. This specific example was done for the uneven counterflow configuration where the cold flow rate is 7 LPM and the hot flow rate is 4 LPM. This was solved by using the NTU effectiveness method which is a heat exchanger analysis method used to predict the heat transfer when outlet temperatures are not known. To complete this calculation, the heat capacity rate for the hot flow, $C_{h,counter}$, and the heat capacity rate for the cold flow, $C_{c,counter}$, needed to be found. These are dependent on the volumetric flowrate, Q, the fluid density, ρ, and the fluid specific heat capacity, C_p, These values explain how much heat each fluid can absorb or release per degree of temperature change.

$$C_{h,x,counter} = Q_{hot} * \rho * C_p \; p.8, Schulze \; [1] \tag{22}$$

$$C_{c,x,counter} = Q_{cold} * \rho * C_p \; p.8 \; Schulze \; [1] \tag{23}$$

After this, the maximum heat capacity rate, C_{max}, and the minimum heat capacity rate, C_{min}, needed to be found to solve for the capacity ratio for the counter flow configuration, $C_{r,x,counter}$. This value explains which fluid limits the heat transfer in the heat exchanger. The fluid with the smaller heat carrying capacity is C_{min} because since the heat capacity rate explains how much heat a fluid can carry per degree, the smaller one can carry less thus limiting the total heat transfer.

$$C_{r,x,counter} = \frac{C_{min}}{C_{max}} \; p.11 \; Schulze \; [8] \tag{24}$$

Next, the number of transfer units, *NTU*, needed to be solved which would then be used to determine the effectiveness, ε, of the heat exchanger relative to the fluid heat capacity. The number of transfer units essentially explains how capable the heat exchanger is at transferring heat and is dependent on the distance along the pipe, x, the experimental heat transfer coefficient, $UA_{experimental}$, and the heat capacity rate for the hot flow, $C_{h,x,counter}$. Using that, the effectiveness, ε, can be solved for which is dependent on the number of transfer units, *NTU*, and the capacity ratio, C_r. After that, the predicted heat transfer, $q_{predicted}$, can be found which is dependent on the heat capacity rate for the hot flow, $C_{h,x,counter}$, the effectiveness, ε, the inlet temperature at the hot flow, $T_{hot,inlet}$, and the inlet temperature at the cold flow, $T_{cold,inlet}$.

$$NTU_{x,counter} = \left(\frac{x * UA_{experimental}}{C_{h,x,counter}}\right) \; p.15 \; Schulze \; [8] \tag{25}$$

$$\varepsilon_{x,counter} = \frac{1 - exp\left[-NTU_{x,counter}\left(1 - C_{r,x,counter}\right)\right]}{1 - C_r * exp\left[-NTU_{x,counter}\left(1 - C_{r,x,counter}\right)\right]} \; p.13 \; Schulze \; [8] \tag{26}$$

$$q_{predicted,x,counter} = C_{h,x,counter} * \varepsilon_{x,counter} * \left(T_{hot,inlet} - T_{cold,inlet}\right) p.7\ Schulze\ [8] \tag{27}$$

Finally, the predicted temperature of the hot fluid at the midpoint location of the pipe, $T_{h,0.5}$, can be found which is dependent on the heat capacity rate for the hot flow, $C_{h,,xcounter}$, the predicted heat transfer, $q_{predicted}$, and the inlet temperature at the hot flow, $T_{hot,inlet}$. The predicted temperature of the cold fluid at the midpoint location of the pipe, $T_{c,0.5}$, can also be found which is dependent on the heat capacity rate for the hot flow, $C_{c,x,counter}$, the predicted heat transfer, $q_{predicted}$, and the inlet temperature at the cold flow, $T_{cold,inlet}$.

$$T_{h,0.5} = T_{hot,inlet} - \left(\frac{q_{predicted,0.5,counter}}{C_{h,x,counter}}\right) p.5\ Schulze\ [9] \tag{28}$$

$$T_{c,0.5} = T_{cold,inlet} - \left(\frac{q_{predicted,0.5,counter}}{C_{c,x,counter}}\right) p.5\ Schulze\ [9] \tag{29}$$

With all the necessary information, the predicted temperatures at the midpoints of the pipe length can then be solved for using the previously listed equations.

$$Q_{hot} = 3.97\ \cancel{LPM} * \frac{10^{-3} m^3}{\cancel{L}} * \frac{1\ \cancel{min}}{60\ s} = 0.0000662\ \frac{m^3}{s}$$

$$\rho = 992.7\ \frac{kg}{m^3}$$

$$C_p = 4179.7\ \frac{J}{Kg * °C}$$

$$C_{h,0.5,counter} = 0.0000662\ \frac{\cancel{m^3}}{s} * 992.7\ \frac{\cancel{kg}}{\cancel{m^3}} * 4179.7\ \frac{J}{\cancel{Kg *}\ °C}$$

$$\therefore\ C_{h,0.5,counter} = 274.5\ \frac{W}{°C}$$

$$Q_{cold} = 6.89\ \cancel{LPM} * \frac{10^{-3} m^3}{\cancel{L}} * \frac{1\ \cancel{min}}{60\ s} = 0.000115\ \frac{m^3}{s}$$

$$\rho = 996.9\ \frac{kg}{m^3}$$

$$C_p = 4181.5\ \frac{J}{Kg * °C}$$

$$C_{c,0.5,counter} = 0.000115\ \frac{\cancel{m^3}}{s} * 996.9\ \frac{\cancel{kg}}{\cancel{m^3}} * 4181.5\ \frac{J}{\cancel{Kg *}\ °C}$$

$$\therefore\ C_{c,0.5,counter} = 478.7\ \frac{W}{°C}$$

$$C_{h,0.5,counter} > C_{c,0.5,counter}$$

$$C_{h,0.5,counter} = C_{min}$$

$$C_{c,0.5,counter} = C_{max}$$

$$C_{r,0.5,counter} = \frac{274.5\ \frac{\cancel{W}}{\cancel{°C}}}{478.7\ \frac{\cancel{W}}{\cancel{°C}}}$$

$$\therefore\ C_{r,0.5,counter} = 0.574$$

x = 0.5

$$UA_{experimental} = 167.3\ \frac{W}{°C}$$

$$NTU_{0.5,counter} = \left(\frac{0.5 * 167.3\ \frac{\cancel{W}}{\cancel{°C}}}{274.5\ \frac{\cancel{W}}{\cancel{°C}}}\right)$$

$$\therefore\ NTU_{0.5,counter} = 0.305$$

$$\varepsilon_{0.5,counter} = \frac{1 - exp\ [-0.305(1 - 0.574)]}{1 - 0.574 * exp\ [-0.305(1 - 0.574)]}$$

$$\therefore\ \varepsilon_{0.5,counter} = 0.245$$

$$T_{hot,inlet} = 42.9\ °C$$

$$T_{cold,inlet} = 27.5\ °C$$

$$q_{predicted,0.5,counter} = 274.5\ \frac{W}{\cancel{°C}} *\ 0.245 * (42.9\ \cancel{°C} - 27.5\ \cancel{°C})$$

$$\therefore\ q_{predicted,0.5,counter} = 1038.7\ \text{W}$$

$$T_{h,0.5} = 42.9\ °C - (\frac{1038.7\ \cancel{W}}{274.5\ \frac{\cancel{W}}{°C}})$$

$$\therefore\ T_{h,0.5} = 39.130\ °C$$

$$T_{c,0.5} = 27.5\ °C - (\frac{1038.7\ \cancel{W}}{478.7\ \frac{\cancel{W}}{°C}})$$

$$\therefore\ T_{c,0.5} = 25.331\ °C$$

The uncertainty in the theoretical overall heat transfer coefficient area product, $\mu_{UA_{theoretica}}$, was found using RSS propagation where the partial derivative of the experimental overall heat transfer coefficient area product equation was taken with respect to each of its variables. Each partial derivative was multiplied by the uncertainty of the variable that each derivative was taken with respect to. Note that when evaluating the uncertainties, the impact of the convective coefficients was significantly greater than that of the geometric quantities and material properties.

$$\mu_{UA_{theoretica}} = \sqrt{\begin{array}{c}\left(\mu_{h_i}\frac{\partial UA_{theoretical}}{\partial h_i}\right)^2 + \left(\mu_{A_i}\frac{\partial UA_{theoretical}}{\partial A_i}\right)^2 + \left(\mu_{D_o}\frac{\partial UA_{theoretical}}{\partial D_o}\right)^2 \\ + \left(\mu_{D_i}\frac{\partial UA_{theoretical}}{\partial D_i}\right)^2 + \left(\mu_{k_p}\frac{\partial UA_{theoretical}}{\partial k_p}\right)^2 + \left(\mu_{L}\frac{\partial UA_{theoretical}}{\partial L}\right)^2 \\ + \left(\mu_{h_o}\frac{\partial UA_{theoretical}}{\partial h_o}\right)^2 + \left(\mu_{A_o}\frac{\partial UA_{theoretical}}{\partial A_o}\right)^2\end{array}} \tag{30}$$

$$\mu_{UA_{theoretical}} = \left(\frac{1}{h_i * A_i} + \frac{\ln\left(\frac{D_o}{D_i}\right)}{2 * \pi * k_p * L} + \frac{1}{h_o * A_o}\right)^{-1} * \sqrt{\begin{array}{c}\mu_{h_i}\frac{1}{h_i^2 * A_i} + \mu_{A_i}\frac{1}{h_i * A_i^2} - \mu_{D_o}\frac{1}{2 * \pi * k_p * L * D_o} \\ + \mu_{D_i}\frac{1}{2 * \pi * k_p * L * D_i} + \mu_{k_p}\frac{\ln\left(\frac{D_o}{D_i}\right)}{2 * \pi * k_p^2 * L} \\ + \mu_{L}\frac{\ln\left(\frac{D_o}{D_i}\right)}{2 * \pi * k_p * L^2} + \mu_{h_o}\frac{1}{h_o^2 * A_o} + \mu_{A_o}\frac{1}{h_o * A_o^2}\end{array}}$$

References:

[1] K. Schulze, *Heat Exchanger 2*. Gainesville, FL: 2026, pp. 3-11.
[2] K. Schulze, *Heat Exchanger 1*. Gainesville, FL: 2026, pp. 9-15.
[3] The Engineering ToolBox, "Water - Density, Specific Weight and Thermal Expansion Coefficient," The Engineering Toolbox, 2003. https://www.engineeringtoolbox.com/water-density-specific-weight-d_595.html
[4] K. Schulze, *Proteus 200_flowmeter_Data_Sheet.* Gainesville, FL: 2026, p. 3
[5] R. W. Fox and J. W. Mitchell, *Fox and McDonald's introduction to fluid mechanics*. Hoboken: Wiley, 2019.
[6] J. H. Lienhard IV and J. H. Lienhard V, *A Heat Transfer Textbook*, 6th ed. Cambridge, MA, USA: Phlogiston Press, 2024. [Online]. Available: https://ahtt.mit.edu/
[7] T. L. Bergman, A. S. Lavine, F. P. Incropera, and D. P. DeWitt, *Fundamentals of Heat and Mass Transfer*, 8th ed. Hoboken, NJ, USA: Wiley, 2017.
[8] K. Schulze, *Heat Exchanger 3*. Gainesville, FL: 2026, pp. 7-15.
[9] K. Schulze, *Heat Exchanger 4*. Gainesville, FL: 2026, p. 5.

Table 5: Lab Report Participation Log

Name	Date	Hours Worked	Description of Tasks Performed
Bryce Addeo	3/1/2026	1	Report formatting, 1st paragraph cover letter, table 1
Matthew Strogach	3/5/2026	0.5	Input experimental temperature and std dev data into charts
Bryce Addeo	3/6/2026	1	LMTD calculations from temperature data
Bryce Addeo	3/8/2026	2.5	Calculations for q, UA, Re, Pr, Nu, & h from data
Bryce Addeo	3/8/2026	1.5	Calculations for UAtheor, Rf, and T at x/L = 0.5
Matthew Strogach	3/9/2026	3	Format and enter values into table 3, LMTD uncertainty calculation, LMTD sample calculation and sample uncertainty calculation
Matthew Strogach	3/9/2026	1	Heat transfer rate and q(avg) uncertainty
Matthew Strogach	3/9/2026	1	Fouling factor sample calculation and table 3 description
Bryce Addeo	3/9/2026	1.5	Completed 4 graphs for predicted vs measured T's
Sean Muller	3/10/2026	2	UA experimental uncertainty calculation
Matthew Strogach	3/10/2026	1.5	Heat transfer rate sample calculation and uncertainty sample calculation
Matthew Strogach	3/10/2026	1	Made corrections to paper at office hours
Bryce Addeo	3/10/2026	1	Made corrections to paper at office hours
Bryce Addeo	3/10/2026	1	Table 1 description
Matthew Strogach	3/11/2026	2	Sample calculations for predicted T at x=½, NTU, cc, ch, cr, effectiveness, and q
Bryce Addeo	3/11/2026	2	Cover letter 2nd paragraph and Table 2 + description
Bryce Addeo	3/12/2026	2.5	Cover letter 3rd paragraph, Fig. 1,2,3,4 descriptions
Sean Muller	3/12/2026	4	UA theoretical uncertainty calculations
Bryce Addeo	3/13/2026	1	Cover letter 4th paragraph & finalized cover letter
Matthew Strogach	3/13/2026	1	Fouling factor uncertainty
Sean Muller	3/13/2026	10	UA table, sample calculations associated with UA theoretical, sources for equations
Bryce Addeo	3/13/2026	2.5	Proofreading and making corrections to report
	Total Hours:	44.5	

Total Hours per Group Member: Bryce: 17.5 hrs, Matt: 11 hrs, Sean: 16 hrs